Heart Diseases Medical Terminology Search Engine

Monir, Thong

Outline

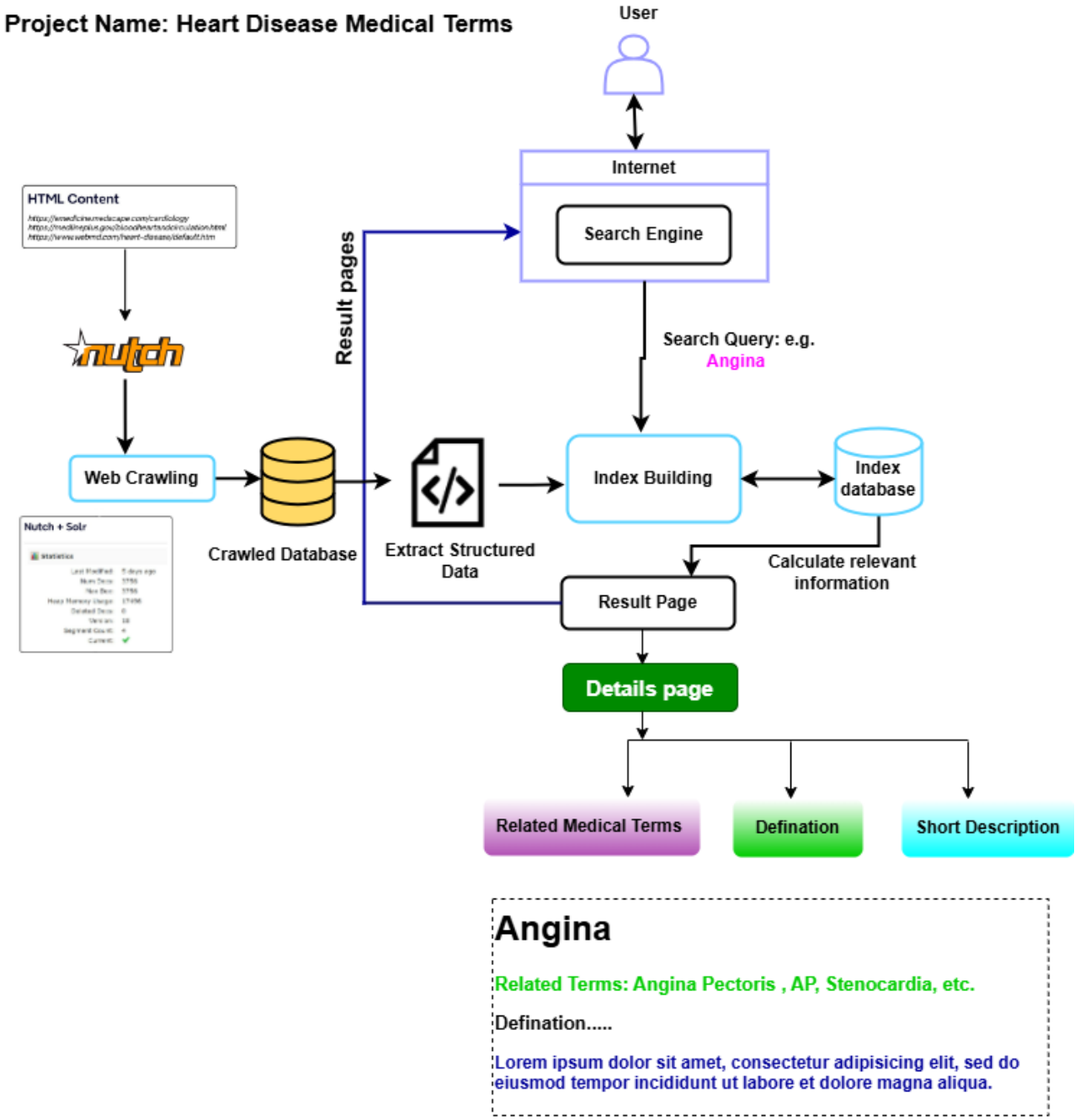
- Introduction
- Architecture
- Crawling with Apache Nutch
- Features profiling
- Extract to structured data
- Matchmaking and ranking
- Implementation and Result

Introduction

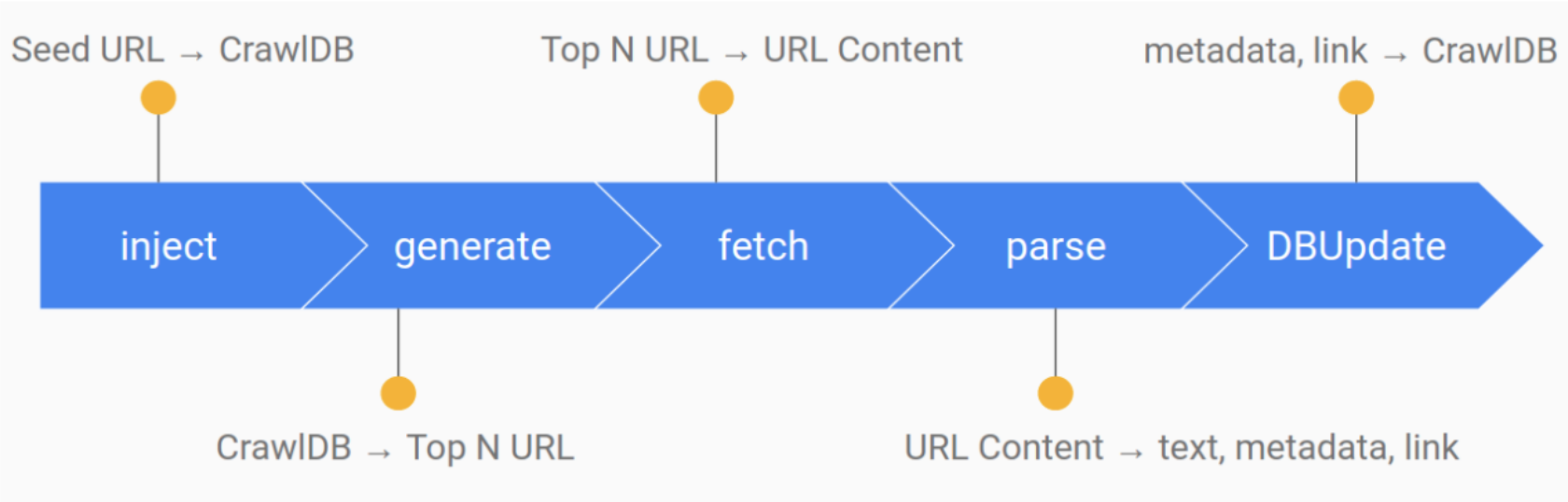
Why users need Medical Terminology Search Engine?

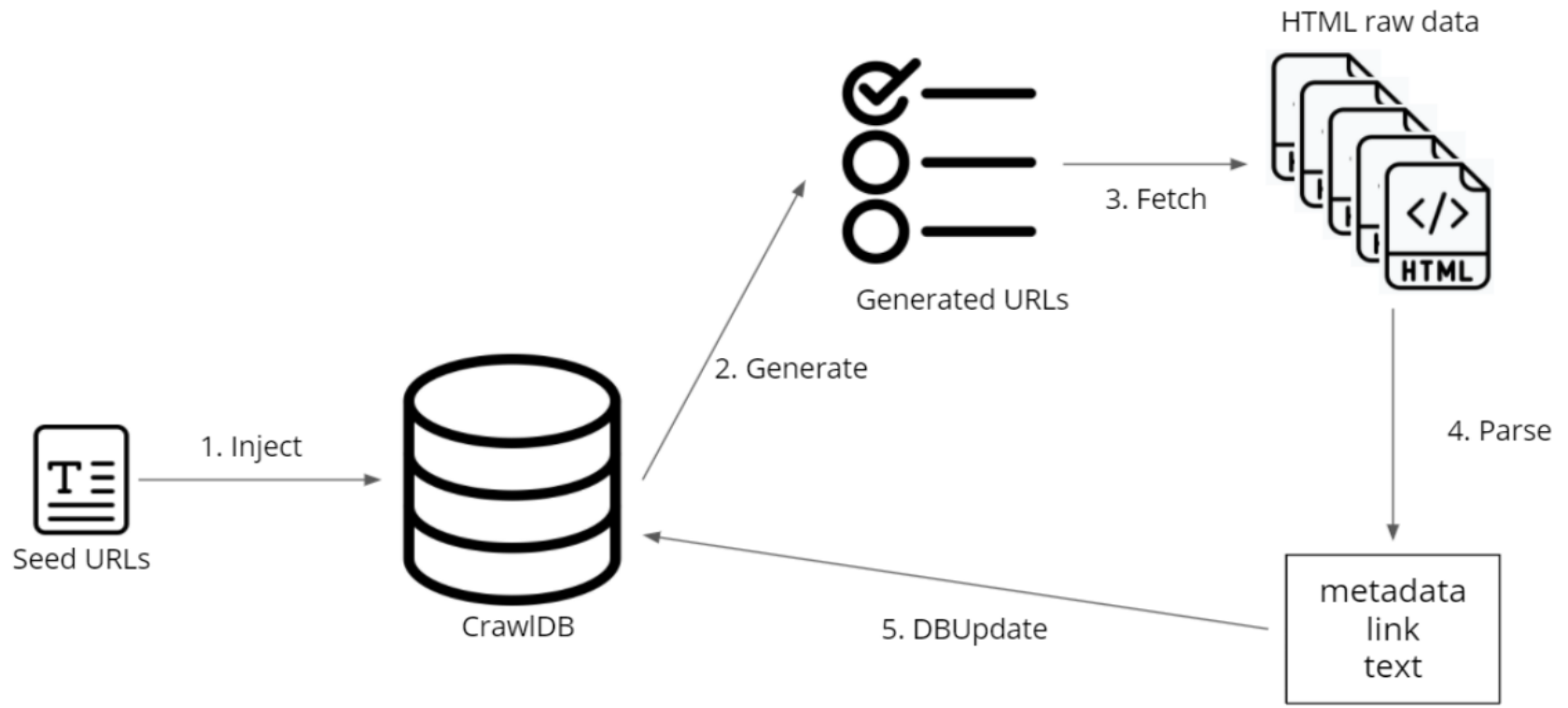
- Detailing medical term with related terms and description all together.
- Providing accurate, reliable information from trusted sources.
- Connecting related information for better understanding.

Architecture



Web Crawling with Apache Nutch





Crawling Process

Data Sources

- seed.txt

```
https://emedicine.medscape.com/cardiology
https://medlineplus.gov/bloodheartandcirculation.html
https://www.webmd.com/heart-disease/default.htm
```

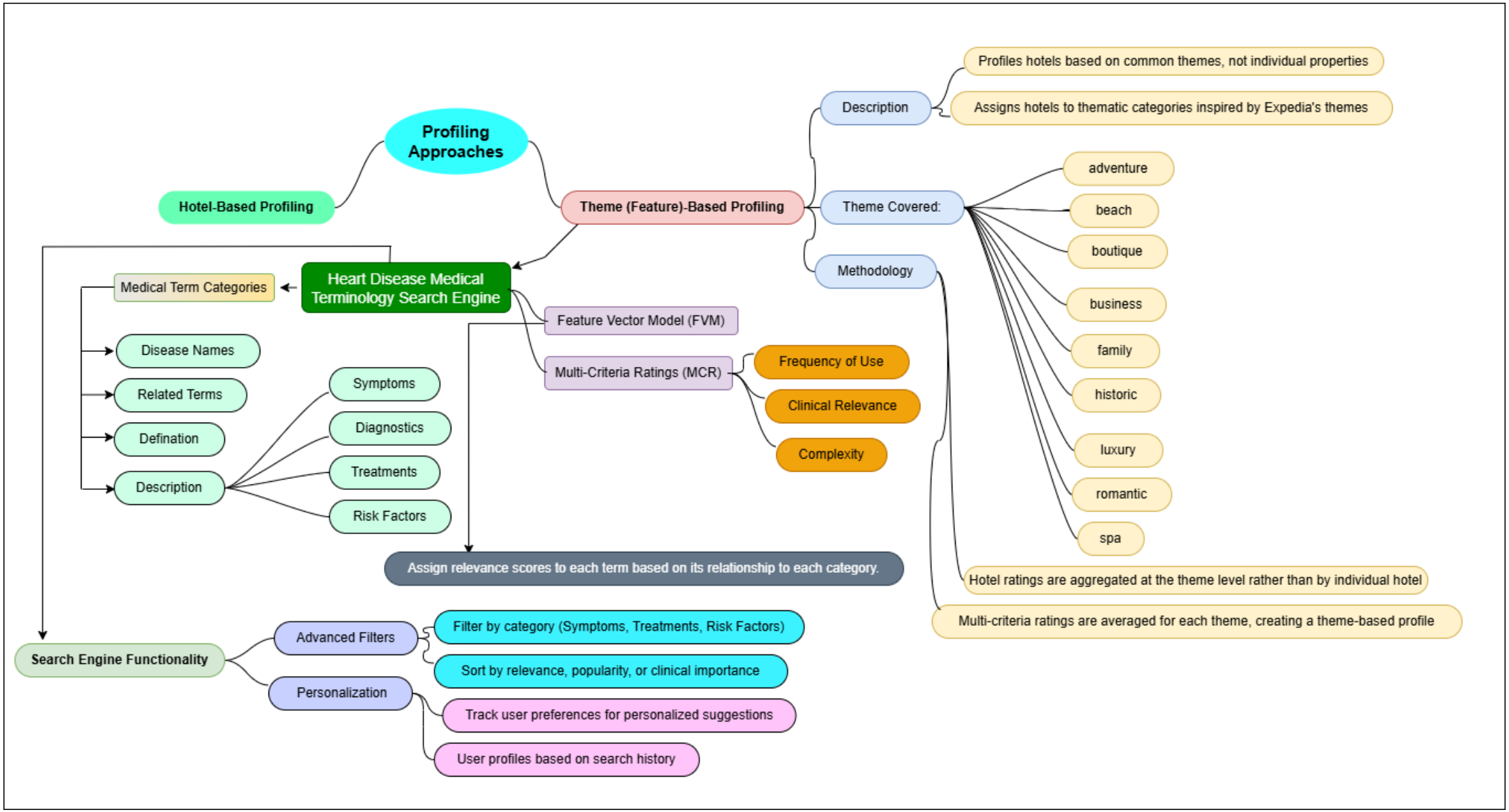
Crawled Database

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If you log out, you will be required to enter your username and password the next time you visit. Log out\nCancel\nDrugs and Diseases\nArticles on\nCardiology\nChange\nShow articles :\nBy Category\nAlphabetically\nExpand All\nArrhythmias\nAccelerated Idioventricular Rhythm\nAntidysrhythmic Toxicity\nArrhythmogenic Right Ventricular Dysplasia/Cardiomyopathy (ARVD/ARVC)\nAshman Phenomenon\nAsystole\nAtrial Fibrillation\nAtrial Flutter\nAtrial Tachycardia\nAtrioventricular Block\nAtrioventricular Dissociation\nAtrioventricular Nodal Reentry Tachycardia\nBrugada Syndrome\nCardiac Resynchronization Therapy\nDigitalis Toxicity\nElectrical Alternans\nFirst-Degree Atrioventricular Block\nHoliday Heart Syndrome\nImplantable Loop Recorder (Insertable Cardiac Monitor)\nJunctional Rhythm\nLong QT Syndrome\nLown-Ganong-Levine Syndrome\nMultifocal Atrial Tachycardia\nPacemaker Malfunction\nPacemaker Syndrome\nPacemaker-Mediated Tachycardia\nParoxysmal Supraventricular Tachycardia\nPremature Ventricular Complexes\nPremature Ventricular Contraction in Emergency Medicine\nPulseless Electrical Activity\nSecond-Degree Atrioventricular Block\nSinus Bradycardia\nSinus Node Dysfunction\nSudden Cardiac Death\nThird-Degree Atrioventricular Block (Complete Heart Block)\nTorsade de Pointes\nVentricular Fibrillation\nVentricular Tachycardia\nWolff-Parkinson-White Syndrome\nAtherosclerosis and Risk Factors\nCardiovascular Disease Primary Prevention/Lifestyle Guidelines\nCoronary Artery Atherosclerosis\nCoronary Bare-Metal Stent\nCoronary Drug-Eluting Stent\nHypertensive Heart Disease\nLipid Management Guidelines\nMalignant Hypertension\nMetabolic Syndrome\nMitral Annular Calcification\nPrimary and Secondary Prevention of Coronary Artery Disease\nRenal Artery Stenosis\nRetinal Macroaneurysm\nRisk Factors for Coronary Artery Disease\nCancer and the Heart\nAtrial Myxoma\nBenign Cardiac Tumors\nCardiac Sarcoma\nCarney Complex\nPrimary Cardiac Neoplasms\nCardiovascular Syndromes in Systemic Diseases\nAortitis\nCardiac Amyloidosis\nHypertension\nLoeffler Endocarditis\nCongenital Heart Disease in the Adult\nAortic Coarctation\nAtrial Septal Defect\nCongenitally Corrected Transposition\nCor Triatriatum\nEbstein Anomaly\nEisenmenger Syndrome\nEndocardial Cushion Defects (Atrioventricular Canal Defects, Atrioventricular Septal Defects)\nHolt-Oram Syndrome\nIsolated Coronary Artery Anomalies\nLutembacher Syndrome\nPatent Foramen Ovale\nSinus of Valsalva Aneurysm\nTetralogy of Fallot (TOF) in Adults\nTricuspid Atresia\nWolff-Parkinson-White Syndrome\nCoronary Artery Disease\nAcute Coronary Syndrome\nAngina Pectoris\nCardiac Rehabilitation\nCardiac Syndrome X\nCarotid Sinus Hypersensitivity\nComparison of Coronary Artery Bypass Grafting (CABG) and Percutaneous Coronary Intervention (PCI)\nComplications of Myocardial Infarction\nCoronary Artery Bypass Grafting\nCoronary Artery Disease Imaging\nCoronary Artery Vasospasm\nCoronary CT Angiography\nMetabolic Syndrome\nMyocardial Infarction\nMyocardial Rupture\nPercutaneous Coronary Intervention (PCI)\nPharmacologic Stress Testing\nPostinfarction Ventricular Septal Rupture\nPrimary and Secondary Prevention of Coronary Artery Disease\nRight Ventricular Infarction\nRisk Factors for Coronary Artery Disease\nSaphenous Vein Graft Aneurysms\nTreadmill Stress Testing\nUnstable Angina\nElectrophysiology Procedures\nImplantable Loop Recorder (Insertable Cardiac Monitor)\nLibman-Sacks Endocarditis\nPacemakers and Implantable Cardioverter-Defibrillators\nInvasive Diagnostic, Interventional, and Surgical Procedures\nAortoiliac Occlusive Disease\nArteriovenous Fistulas and Malformations\nComparison of Coronary Artery Bypass Grafting (CABG) and Percutaneous Coronary Intervention (PCI)\nEmergency Bedside Thoracotomy\nHeart Transplantation\nHeart-Lung Transplantation\nPediatric Heart Transplantation\nPericardiocentesis\nProsthetic Heart Valves\nSuperior Vena Cava Syndrome (SVCS)\nThoracic Endovascular Aortic Repair (TEVAR)\nTube Thoracostomy\nTube Thoracostomy Management\nVenous Air Embolism\nMyocardial Disease and Cardiomyopathies\nAcute Myocardial Infarction Imaging\nAlcoholic Cardiomyopathy\nCardiac Cirrhosis and Congestive Hepatopathy\nCardiogenic Pulmonary Edema\nCardiogenic Shock\nCocaine-Related Cardiomyopathy\nCor Pulmonale\nDilated Cardiomyopathy (DCM)\nDilated Cardiomyopathy Imaging\nEndomyocardial Fibrosis\nHeart Failure\nHypertrophic Cardiomyopathy\nHypertrophic Cardiomyopathy Imaging\nMyocardial Abscess\nMyocarditis\nMyocarditis Organism-Specific Therapy\nMyocarditis Pathology\nMyopathies\nPathology of Acute Myocardial Infarction\nPediatric Viral Myocarditis\nPeripartum (Postpartum) Cardiomyopathy (PPCM)\nRestrictive Cardiomyopathy\nTakotsubo (Stress) Cardiomyopathy (Broken Heart Syndrome)\nThrombolysis in Myocardial Infarction (TIMI) Score\nPericardial Disease\nAcute Pericarditis\nCardiac Tamponade\nConstrictive Pericarditis\nConstrictive Pericarditis Imaging\nEffusive-Constrictive Pericarditis\nPericardial Effusion\nPericardial Effusion Imaging\nPericardial Fluid Analysis\nPericarditis Empiric Therapy\nPericarditis Organism-Specific Therapy\nPostpericardiotomy Syndrome\nPeripheral Vascular Disease\nAcute Aortic Dissection\nAnkle-Brachial Index (ABI) Measurement\nCarotid Artery Dissection\nDeep Vein Thrombosis Bedside Ultrasonography\nDeep Venous Thrombosis (DVT)\nInferior Vena Cava Filters\nNoncoronary Atherosclerosis\nPeripheral Vascular Disease\nSubclavian Artery Thrombosis\nSubclavian Vein Thrombosis\nTakayasu Arteritis\nThoracic Aneurysm\nThoracic Aortic Aneurysm\nVertebral Artery Dissection\nSpecial Patient Populations\nAnticoagulants and Thrombolytics in Pregnancy\nCardiovascular Disease and Pregnancy\nValvular Heart Disease\nAortic Regurgitation\nAortic Stenosis\nIdiopathic Pulmonary Arterial Hypertension\nInfective Endocarditis\nLibman-Sacks Endocarditis\nLoeffler Endocarditis\nLutembacher Syndrome\nMitral Annular Calcification\nMitral Regurgitation\nMitral Stenosis\nMitral Valve Prolapse\nProsthetic Heart Valves\nPulmonary Arterial Hypertension\nPulmonary Regurgitation (Pulmonic Regurgitation)\nPulmonic Stenosis (Pulmonary Stenosis)\nPulmonic Valvular Stenosis\nRheumatic Fever\nSupravalvular Ring Mitral Stenosis\nTricuspid Atresia\nTricuspid Regurgitation\nTricuspid Stenosis\nAll\nA\nB\nC\nD\nE\nF\nG\nH\nI\nJ\nK\nL

Ln 1, Col 427 | 56,397,650 characters | 100% | Windows (CRLF) | UTF-8

Features profiling

Feature Profiling Mind Map



https://doi.org/10.1016/j.elerap.2019.100832

Features for Medical Terminology Search Engine

- Related Medical Terms
- Definition
- Symptoms
- Diagnosis
- Treatment
- Risk Factors

Generate Related Medical Terms by Snomed API (e.g. Angina)

```python
def getSnomedCode(searchTerm):

    for term in data['items']:
        if searchTerm.lower() == term['term'].lower():
            return(term['concept']['conceptId'])

def getConceptById(id):
    url = baseUrl + '/browser/' + edition + '/' + version + '/concepts/' + id
    response = urlopen_with_header(url).read()
    return(json.loads(response.decode('utf-8')))

def getTerm(concept):
    term = []
    for item in concept['descriptions']:
        if item['active']:
            term.append(item['term'])
    return(term)

originalTerm = 'Angina Pectoris'
id = getSnomedCode(originalTerm)
concept = getConceptById(id)
relatedTerms = getTerm(concept)
print(relatedTerms)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python

```
PS C:\Users\monir\Medical Terminology> & C:/Users/monir/anaconda3/python.exe "c:/Users/monir/Medical Terminology/final_snomed.py"
['Angina', 'Stenocardia', 'Anginal syndrome', 'Angina pectoris', 'Ischemic heart disease - angina', 'AP - Angina pectoris', 'Angina (disorder)', 'Cardiac angina', 'Ischaemic heart disease - angina']
```

Extract to structured data

Natural Language Processing (NLP) Techniques

- Core NLP
- Advanced NLP Techniques

Extract “term” from “title” in crawled data

- Use Named Entity Recognition (NER) NLP Technique in SpaCy's biomedical model (***“en_ner_bc5cdr_md”***) to extract topic keywords (medical terms) from the titles.
- The keywords with label “DISEASE” will be extracted as a medical terminology.

```python
# Load SciSpaCy's medical model
nlp = spacy.load("en_ner_bc5cdr_md")

# NLP Function Extract Terms
def extract_medical_keywords(title):
    doc = nlp(title)
    medical_terms = [ent.text for ent in doc.ents if ent.label_ == "DISEASE"]
    return medical_terms
```

Extract “definition” feature

- **Sentence Segmentation:**

  Use SpaCy's biomedical model (***“en_core_sci_sm”***) to split the *medical content* into sentences.

- **Question-Answering (QA) Model:**

  Use the QA model (***“deepset/roberta-base-squad2”***) to answer the question: What is <*term*>? to detect sentences that define the terms.

```python
import spacy
from transformers import AutoModelForQuestionAnswering, AutoTokenizer, pipeline

# Load NLP model (scispaCy for medical texts)
nlp = spacy.load("en_core_sci_sm")

# Load QA model & tokenizer
model_name = "deepset/roberta-base-squad2"
model = AutoModelForQuestionAnswering.from_pretrained(model_name)
tokenizer = AutoTokenizer.from_pretrained(model_name)

# Create a question-answering pipeline
qa_pipeline = pipeline('question-answering', model=model, tokenizer=tokenizer)
```

Extracting Medical Information (Description)

Objective: The goal is to extract key medical information (Symptoms, Diagnosis, Treatment, and Risk Factors) from a large corpus of medical text and summarize it.

Key Components:

- **SpaCy**: Used for natural language processing (NLP) and entity extraction from the medical text.
- **Hugging Face Transformers**: Utilized for text summarization using the ***facebook/bart-large-cnn*** model.
- **Chunking and Tokenization**: Text is tokenized into manageable chunks to fit within the model's token limit (1024 tokens), ensuring it can be processed without errors.

```
3   import spacy
4   import csv
5   from transformers import pipeline
6
7   # Load SciSpaCy for medical term extraction
8   nlp = spacy.load("en_core_sci_sm")
9
10  # Load the summarization pipeline
11  summarizer = pipeline("summarization", model="facebook/bart-large-cnn")
```

Description Extraction and Summarization

1. Preprocessing:

- The JSON file is read and sanitized to remove invalid characters (like newlines and control characters).
- The data is parsed into a Python dictionary.

2. Text Chunking:

- Large text blocks are split into smaller chunks based on token length using the tokenizer from Hugging Face.
- Each chunk is ensured to stay within the model's token limit of 1024 tokens to avoid errors.

3. Keyword-based Extraction:

- Keywords like "symptom", "treatment", "diagnosis", and "risk" are used to filter and extract relevant sentences from the content.

4. Summarization:

- Each relevant section is summarized using the ***facebook/bart-large-cnn*** model.
- The summarizer ensures that no chunk exceeds the token limit while generating meaningful, concise summaries.

Output Example: Extracted Information

- The code successfully extracts and categorizes medical information under four headings: **Symptoms**, **Diagnosis**, **Treatment**, and **Risk Factors**.
- For example, the term *Angina Pectoris* is summarized with related symptoms, diagnostic methods, treatments, and common risk factors.

1	Term	Symptoms	Diagnosis	Treatment	Risk Factors
2	Angina Pectoris	You must confirm that you would like to log out of Medscape. If you log out, you will be required to enter your username and password the next time you visit. Angina pectoris is more often presenting the symptom of coronary artery disease in women than in men. Important prognostic indicators in patients with angina pectoris include LV function, severity and location of atherosclerotic lesions, and response of symptoms to medical treatment. Ivabradine in Combination with Metoprolol Improves Symptoms and Quality of Life in Patients with St	Chest radiography: Usually normal in angina pectoris but may show cardiomegaly in patients with previous MI, ischemic cardiomyopathy, pericardial effusion, or acute pulmonary edema. Graded exercise stress testing: This is the most	Angina pectoris is the result of myocardial ischemia caused by an imbalance between my heart's blood supply and oxygen demand. The study is a prospective, open-label, multicenter, safety feasibility first-in-man study. All material on this website is protected by copyright, Copyright Â© 1994-2024 by WebMD LLC. Medscape has a Terms of Use that includes the use of cookies. Use the Daily Discussion to help people with reading comprehension and vocabulary.	The 2006 overall preliminary death rate from cardiovascular disease was 262.9. Complications of angina pectoris include unstable angina, MI, and death.

Contents of "terms_and_descriptions.csv"

Extracted Database


```
{
    "term": "atherosclerosis",
    "definition": " What is atherosclerosis? Atherosclerosis is a condition in which plaque builds up inside your arteries.
    "symptoms": "In the early stages, atherosclerosis often does not cause any symptoms. For men, erectile dysfunction (ED)
    "diagnosis": "How is atherosclerosis diagnosed?  How is a person diagnosed with Atherosclerosis? How is the disease diag
    "treatment": "To find out if you have atherosclerosis, your provider will ask about your medical and family health histo
    "risk_factors": "Atherosclerosis can cause other health problems, or complications. Treatment for atherosclerosis or its
},
```

Matchmaking and Ranking

Ranking by Cosine Similarity

- **TF-IDF (Term Frequency-Inverse Document Frequency)**:
  - Measures the importance of words in a document relative to a collection of documents.
  - **Term Frequency (TF)**: How often a word appears in a document.
  - **Inverse Document Frequency (IDF)**: Penalizes common words by assigning lower importance to words that appear in many documents.
  - *TfidfVectorizer* function converts the *term* and *definitions* into numerical TF-IDF vectors.
- **Cosine Similarity:**
  - Measures the cosine of the angle between two vectors in a multi-dimensional space. It quantifies the similarity between two pieces of text.
  - The cosine similarity between the *term* and each *definitions* is computed by comparing their TF-IDF vectors using *cosine_similarity* function.

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

Ranking Example: Angina

```python
# Main function extract definition
query = f"What is {term}?"
candidates = []
for item in sample_data:
    content = item['content']
    content = re.sub(r'\n', ' ', content)
    doc = nlp(content)
    sentences = [sent.text for sent in doc.sents]
    candidate = ""
    for sentence in sentences:
        result = qa_pipeline(question=query, context=sentence)
        if result['score'] > 0.3:  # Filter by confidence score
            candidate = candidate + " " + sentence
    if candidate != "":
        candidates.append(candidate)
```

Python

```python
# Ranking by Cosine Similarity
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform([term] + candidates)
similarities = cosine_similarity(tfidf_matrix[0:1], tfidf_matrix[1:]).flatten()
ranked_candidates = sorted(
    enumerate(candidates, 1), key=lambda x: similarities[x[0] - 1], reverse=True
)
print("Ranked Definition of", term,":")
for rank, (index, sentence) in enumerate(ranked_candidates, 1):
    print(f"{rank}. [Similarity: {similarities[index-1]:.4f}]{sentence}")
```

✓ 0.0s Python

```
Ranked Definition of angina :
1. [Similarity: 0.4942] Unstable angina is a condition in which your heart doesn't get enough blood flow and oxygen. Angina is a type of chest discomfort due to poor blood flow through the blood vessel
2. [Similarity: 0.4201] Angina is a symptom of coronary artery disease (CAD), the most common heart disease . There are three types of angina: Stable angina is the most common type. Unstable angina is
3. [Similarity: 0.3927] Angina is chest pain that happens because there isn't enough blood going to part of your heart. It's sometimes called angina pectoris or ischemic chest pain. Angina Types There
4. [Similarity: 0.3195] Read on to learn more about unstable angina and how to manage your symptoms.   The lack of blood flow to the heart causes angina — a symptom of coronary artery disease.   Stable
5. [Similarity: 0.3092] The first English-written account of recurrent angina pectoris was by English nobleman Edward Hyde, Earl of Clarendon. 1 ] The first description of angina as a medical disorder
6. [Similarity: 0.2696] Stable angina is chest pain or discomfort that most often occurs with activity or emotional stress. Angina is due to poor blood flow through the blood vessels in the heart calle
7. [Similarity: 0.2652] Unstable Angina. View Media Gallery Signs and symptoms With unstable angina, symptoms may (1) occur at rest; (2) become more frequent, severe, or prolonged than the usual patter
8. [Similarity: 0.1950] Abdominal Angina Overview Practice Essentials Pathophysiology and Etiology Epidemiology Prognosis Show All Presentation History Physical Examination Show All Workup Laboratory S
9. [Similarity: 0.1940] Signs and symptoms Patients should be asked about the frequency of angina, severity of pain, and number of nitroglycerin pills used during episodes. Diagnosis Diagnostic studies
10. [Similarity: 0.1898] Badham first used the currently accepted term angina bullosa hemorrhagica in 1967. The lesions may be confused with other more serious disorders (eg, mucous membrane pemphigoid
```

Ranking Example: Heart Attack

```python
# Main function extract definition
query = f"What is {term}?"
candidates = []
for item in sample_data:
    content = item['content']
    content = re.sub(r'\n', ' ', content)
    doc = nlp(content)
    sentences = [sent.text for sent in doc.sents]
    candidate = ""
    for sentence in sentences:
        result = qa_pipeline(question=query, context=sentence)
        if result['score'] > 0.3:  # Filter by confidence score
            candidate = candidate + " " + sentence
    if candidate != "":
        candidates.append(candidate)
```

✓ 32.7s Python

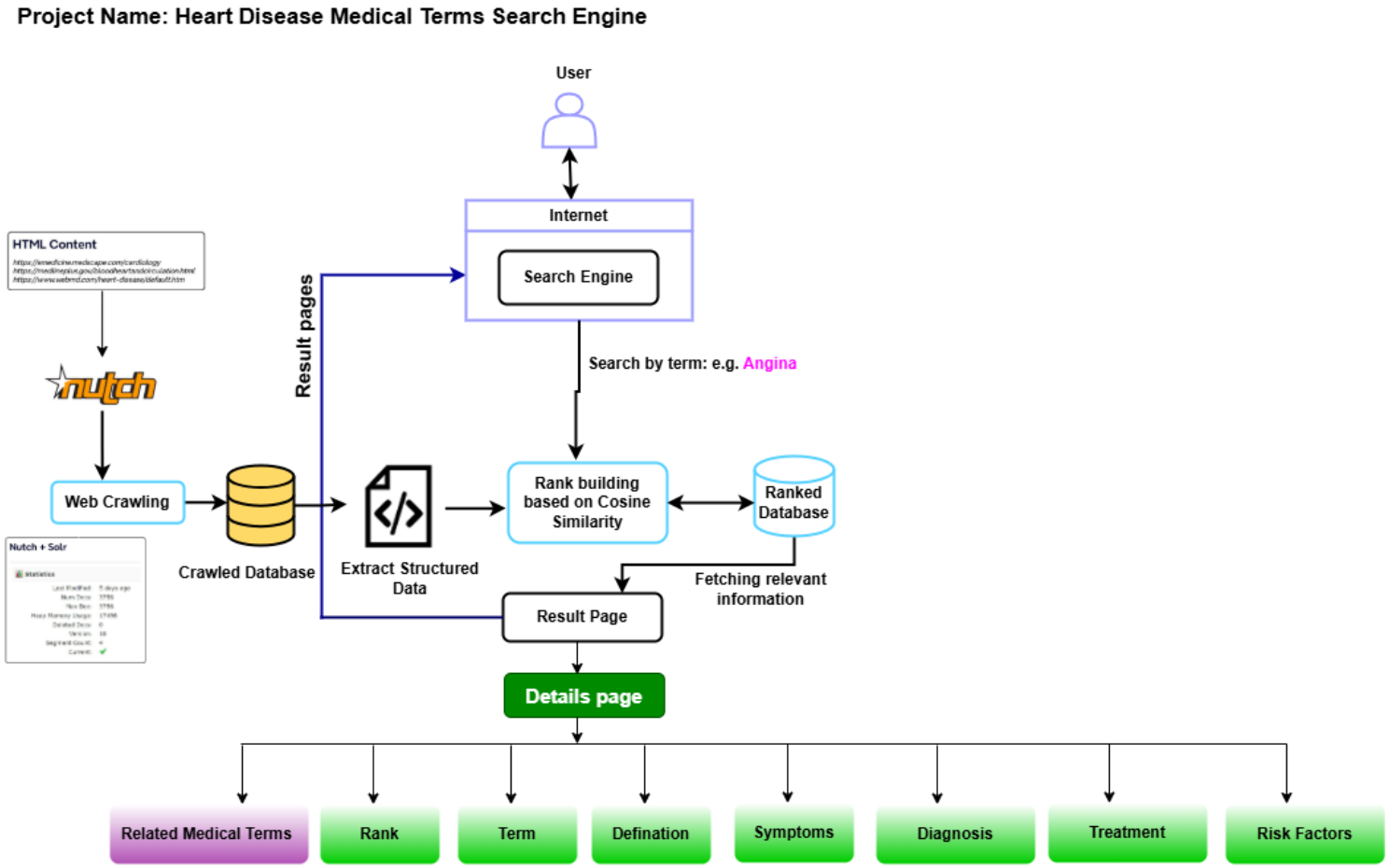
```python
# Ranking by Cosine Similarity
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform([term] + candidates)
similarities = cosine_similarity(tfidf_matrix[0:1], tfidf_matrix[1:]).flatten()
ranked_candidates = sorted(
    enumerate(candidates, 1), key=lambda x: similarities[x[0] - 1], reverse=True
)
print("Ranked Definition of", term,":")
for rank, (index, sentence) in enumerate(ranked_candidates, 1):
    print(f"{rank}. [Similarity: {similarities[index-1]:.4f}]{sentence}")
```

✓ 0.0s Python

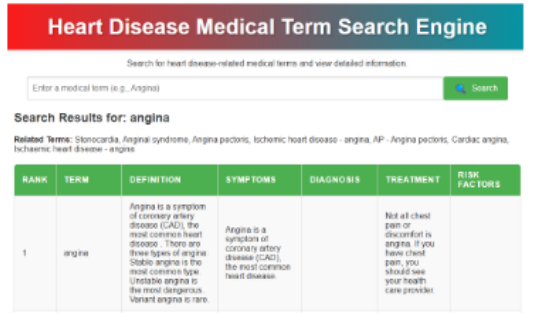
```
Ranked Definition of heart attack :
1. [Similarity: 0.5223] Photo Credit: BSIP / Medical Images) Heart attacks are also called myocardial infarctions (MIs). “ A heart attack feels like intense chest pain, the kind where someone is squeez
2. [Similarity: 0.4433] A heart attack is a medical emergency. Causes A heart attack occurs when the blood flow that carries oxygen to the heart is blocked.
3. [Similarity: 0.4112] Most heart attacks are caused by a blood clot that blocks one of the coronary arteries. A heart attack may occur when: A disruption in the plaque occurs. Symptoms A heart attack
4. [Similarity: 0.3182] A heart attack happens when blood flow to the heart suddenly becomes blocked. That's why it's important to know the symptoms of a heart attack and call 911 if you or someone els
5. [Similarity: 0.2493] A heart attack can occur if a blood clot stops the flow of blood through one of these arteries. This type of heart attack is called an ST segment elevation myocardial infarction
6. [Similarity: 0.2479] Now, a top U.S. panel of health experts has published recommendations that aim to limit people’s use of daily aspirin to prevent a first heart attack or stroke. People 60 and ol
7. [Similarity: 0.1282] The chain of survival is a series of actions that, when done in sequence, will give a person having a heart attack the best chance of survival.
```

Implementation and Result

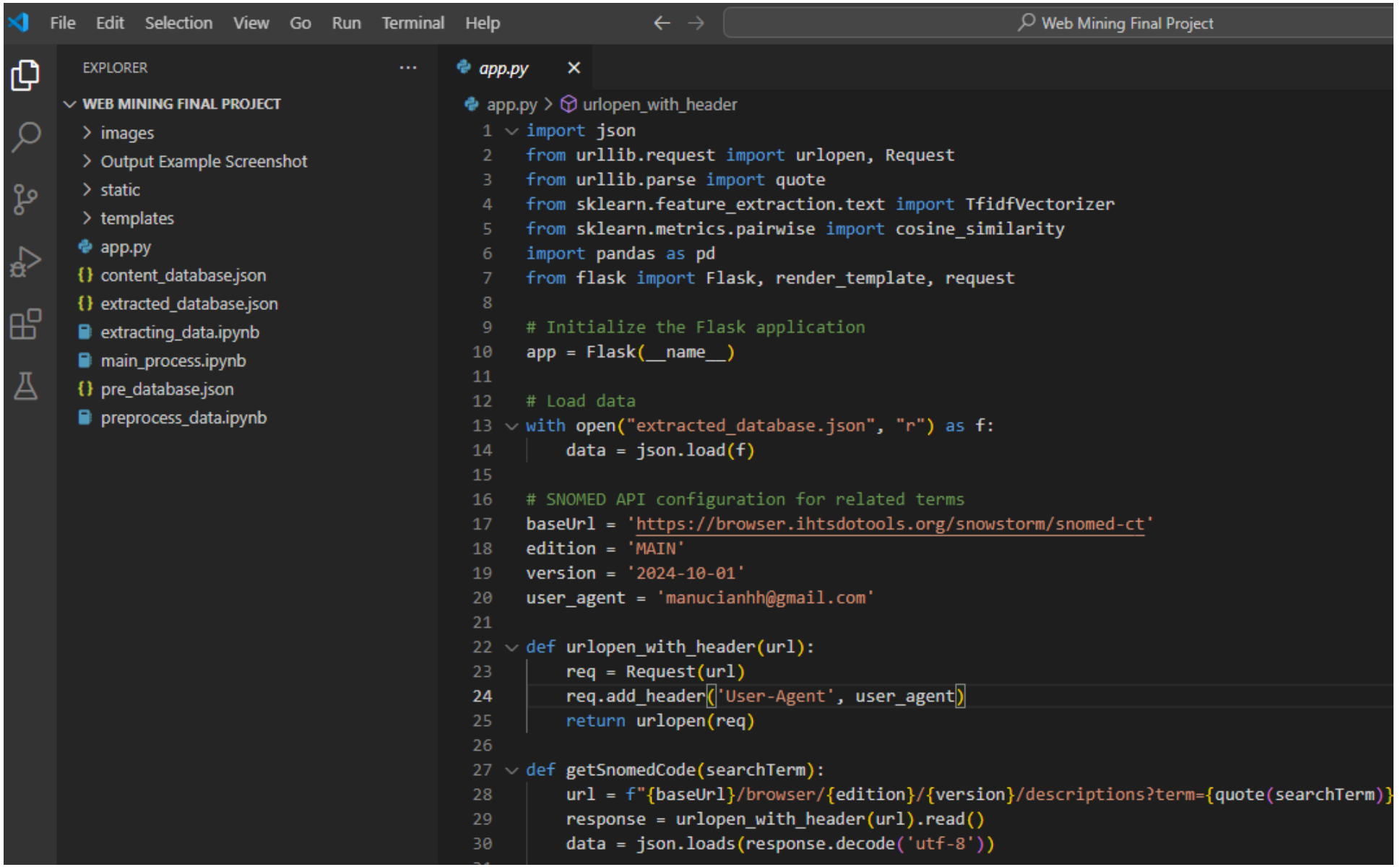
Updated Architecture



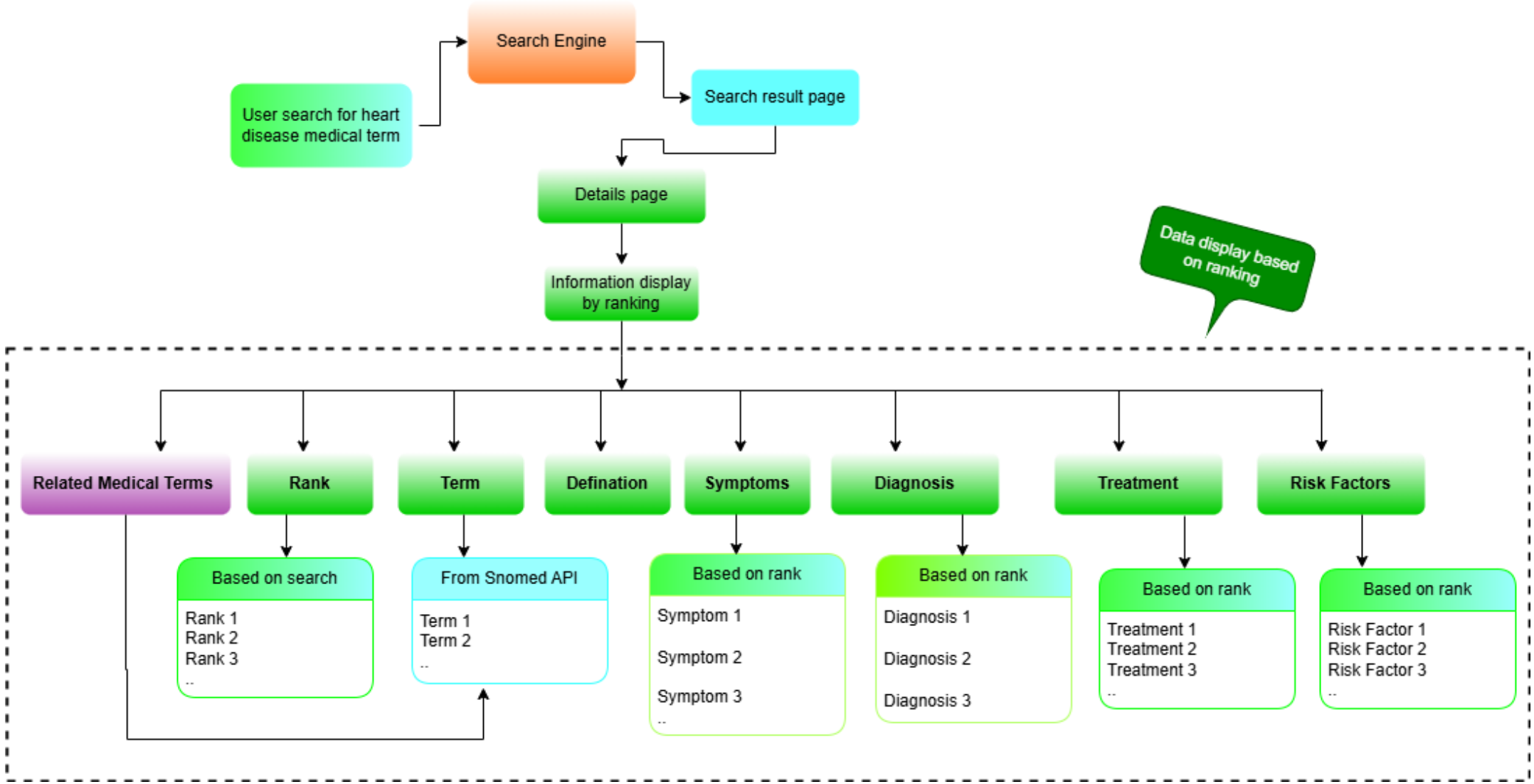
Output Example:



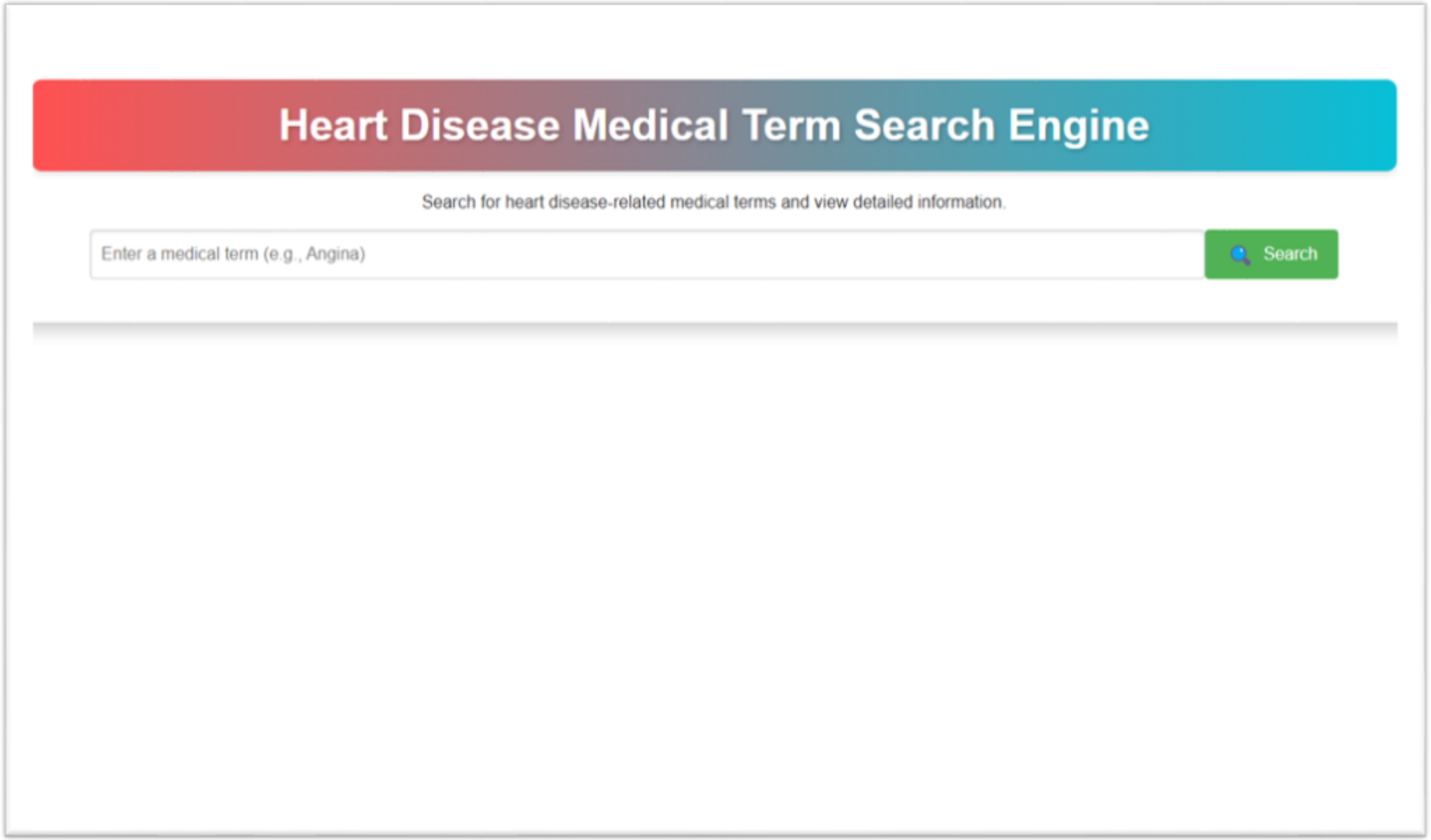
Final Project Code



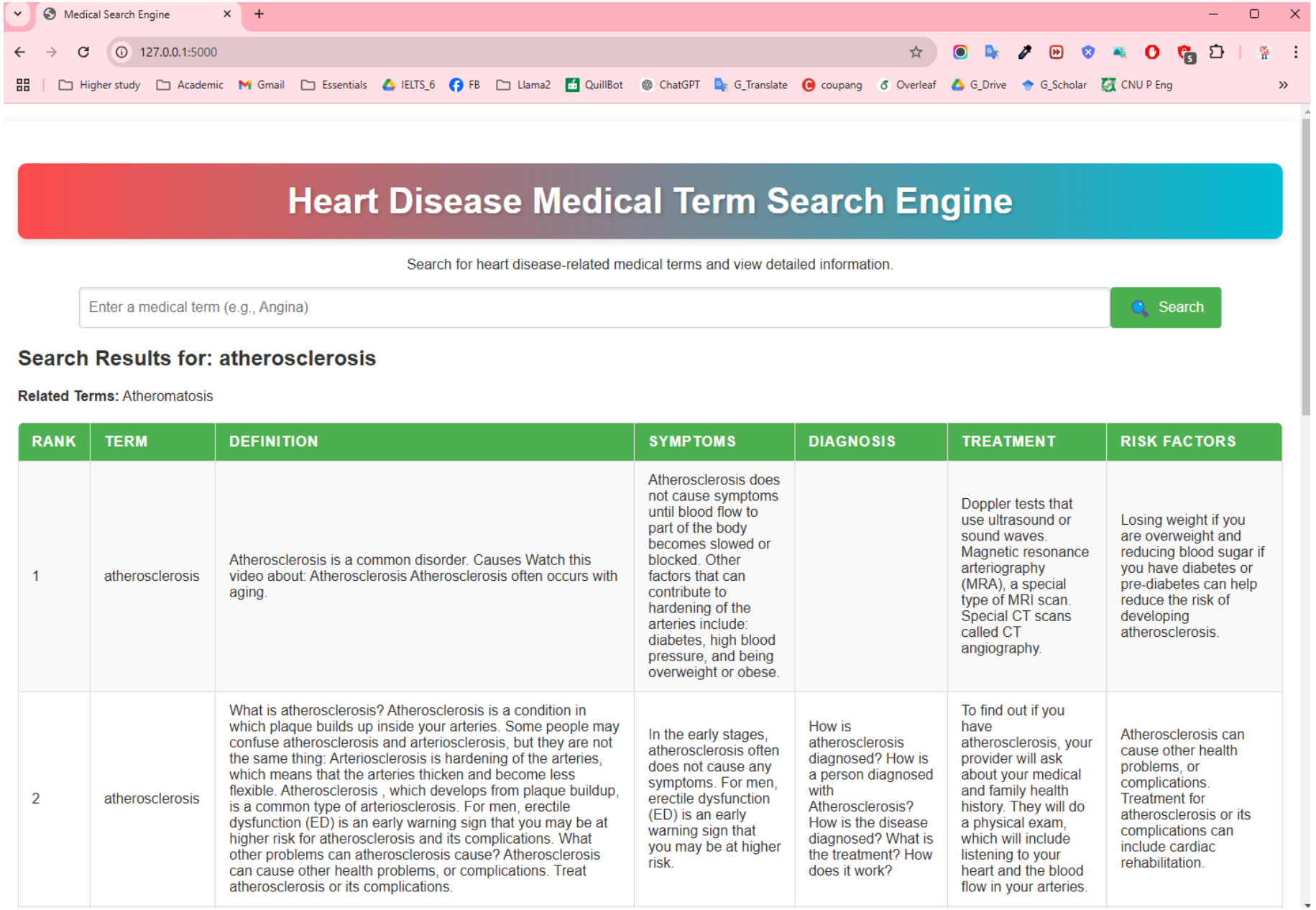
User Case Scenario



User Interface



Output Example: Atherosclerosis



RANK	TERM	DEFINITION	SYMPTOMS	DIAGNOSIS	TREATMENT	RISK FACTORS
1	atherosclerosis	Atherosclerosis is a common disorder. Causes Watch this video about: Atherosclerosis Atherosclerosis often occurs with aging.	Atherosclerosis does not cause symptoms until blood flow to part of the body becomes slowed or blocked. Other factors that can contribute to hardening of the arteries include: diabetes, high blood pressure, and being overweight or obese.		Doppler tests that use ultrasound or sound waves. Magnetic resonance arteriography (MRA), a special type of MRI scan. Special CT scans called CT angiography.	Losing weight if you are overweight and reducing blood sugar if you have diabetes or pre-diabetes can help reduce the risk of developing atherosclerosis.
2	atherosclerosis	What is atherosclerosis? Atherosclerosis is a condition in which plaque builds up inside your arteries. Some people may confuse atherosclerosis and arteriosclerosis, but they are not the same thing: Arteriosclerosis is hardening of the arteries, which means that the arteries thicken and become less flexible. Atherosclerosis , which develops from plaque buildup, is a common type of arteriosclerosis. For men, erectile dysfunction (ED) is an early warning sign that you may be at higher risk for atherosclerosis and its complications. What other problems can atherosclerosis cause? Atherosclerosis can cause other health problems, or complications. Treat atherosclerosis or its complications.	In the early stages, atherosclerosis often does not cause any symptoms. For men, erectile dysfunction (ED) is an early warning sign that you may be at higher risk.	How is atherosclerosis diagnosed? How is a person diagnosed with Atherosclerosis? How is the disease diagnosed? What is the treatment? How does it work?	To find out if you have atherosclerosis, your provider will ask about your medical and family health history. They will do a physical exam, which will include listening to your heart and the blood flow in your arteries.	Atherosclerosis can cause other health problems, or complications. Treatment for atherosclerosis or its complications can include cardiac rehabilitation.

Result comparison with other webpages

https://www.nhs.uk/conditions/atherosclerosis/

Atherosclerosis is where your arteries become narrowed, making it difficult for blood to flow through them. It increases your risk of heart attack and stroke.

https://www.nhlbi.nih.gov/health/atherosclerosis

Atherosclerosis is a common disease that develops when a sticky plaque called plaque builds up in your arteries. Diseases linked to atherosclerosis are the leading cause of death in the United States and the world.

Atherosclerosis develops slowly as cholesterol, fat, blood cells, and other substances in your blood form plaque in the walls of arteries. As plaque builds up, your arteries narrow. This reduces the supply of oxygen-rich blood to tissues and organs in the body.

https://en.wikipedia.org/wiki/Atherosclerosis

Atherosclerosis[a] is a pattern of the disease arteriosclerosis,[8] characterized by development of abnormalities called lesions in walls of arteries. This is a chronic inflammatory disease involving many different cell types and driven by elevated levels of cholesterol in the blood.[9] These lesions may lead to narrowing of the arterial walls due to buildup of atheromatous plaques.[10][11] At the onset there are usually no symptoms,

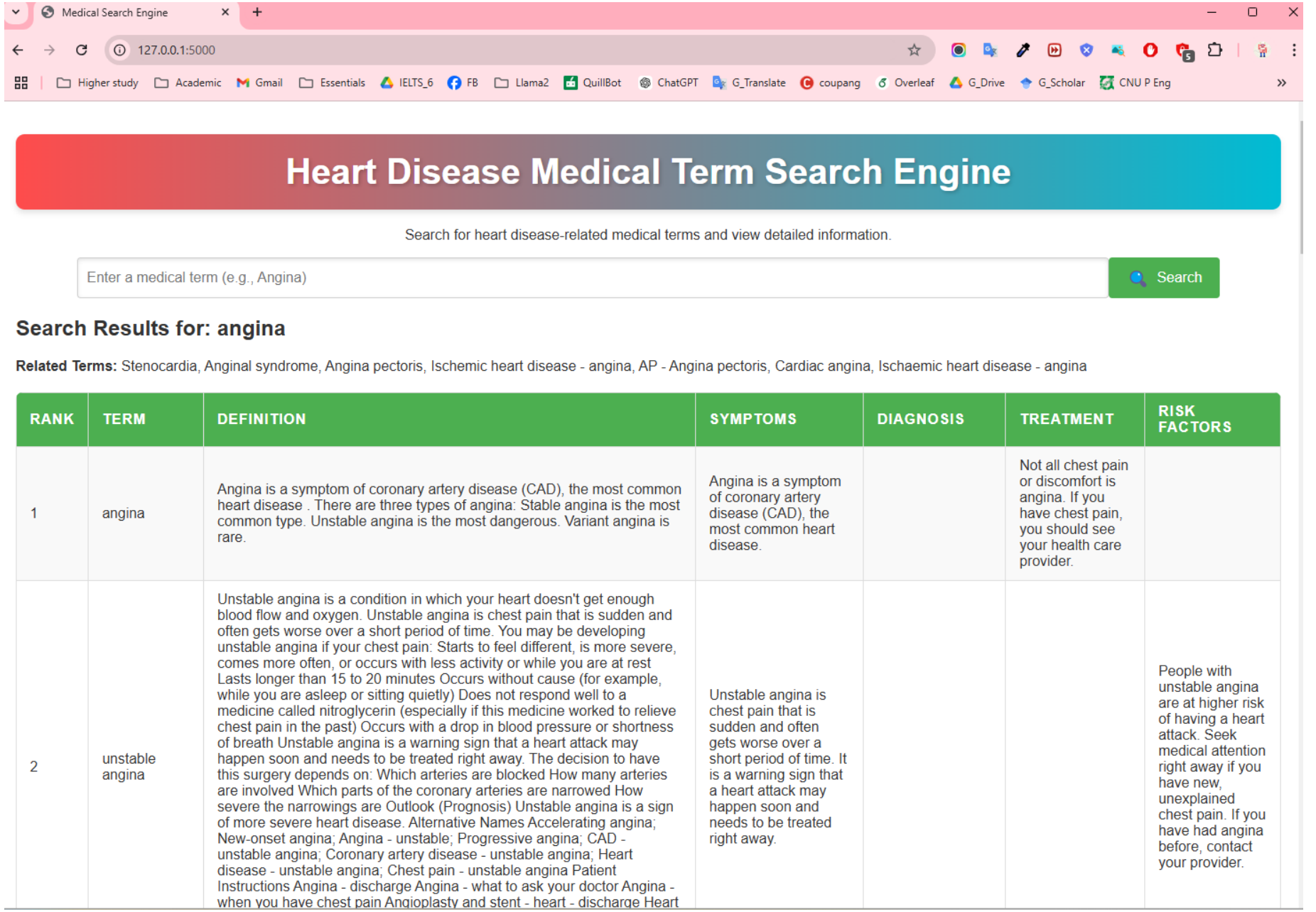
Search Results for: atherosclerosis

Related Terms: Atheromatosis

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Plaque often starts to form during childhood and gets worse with age. Risk factors include high blood cholesterol, unhealthy lifestyle habits, and your genes.

Output Example: Angina



RANK	TERM	DEFINITION	SYMPTOMS	DIAGNOSIS	TREATMENT	RISK FACTORS
1	angina	Angina is a symptom of coronary artery disease (CAD), the most common heart disease . There are three types of angina: Stable angina is the most common type. Unstable angina is the most dangerous. Variant angina is rare.	Angina is a symptom of coronary artery disease (CAD), the most common heart disease.		Not all chest pain or discomfort is angina. If you have chest pain, you should see your health care provider.	
2	unstable angina	Unstable angina is a condition in which your heart doesn't get enough blood flow and oxygen. Unstable angina is chest pain that is sudden and often gets worse over a short period of time. You may be developing unstable angina if your chest pain: Starts to feel different, is more severe, comes more often, or occurs with less activity or while you are at rest Lasts longer than 15 to 20 minutes Occurs without cause (for example, while you are asleep or sitting quietly) Does not respond well to a medicine called nitroglycerin (especially if this medicine worked to relieve chest pain in the past) Occurs with a drop in blood pressure or shortness of breath Unstable angina is a warning sign that a heart attack may happen soon and needs to be treated right away. The decision to have this surgery depends on: Which arteries are blocked How many arteries are involved Which parts of the coronary arteries are narrowed How severe the narrowings are Outlook (Prognosis) Unstable angina is a sign of more severe heart disease. Alternative Names Accelerating angina; New-onset angina; Angina - unstable; Progressive angina; CAD - unstable angina; Coronary artery disease - unstable angina; Heart disease - unstable angina; Chest pain - unstable angina Patient Instructions Angina - discharge Angina - what to ask your doctor Angina - when you have chest pain Angioplasty and stent - heart - discharge Heart	Unstable angina is chest pain that is sudden and often gets worse over a short period of time. It is a warning sign that a heart attack may happen soon and needs to be treated right away.			People with unstable angina are at higher risk of having a heart attack. Seek medical attention right away if you have new, unexplained chest pain. If you have had angina before, contact your provider.

Result comparison with other webpages

https://www.nhs.uk/conditions/angina/

Angina is chest pain caused by reduced blood flow to the heart muscles. It's not usually life threatening, but it's a warning sign that you could be at risk of a heart attack or stroke.

https://www.nhlbi.nih.gov/health/angina

Angina, also known as angina pectoris, is chest pain or discomfort that occurs when part of your heart muscle does not get enough oxygen-rich blood. It is a common symptom of coronary heart disease, which develops when the arteries of the heart become partially or totally blocked.

https://en.wikipedia.org/wiki/Angina

Angina, also known as **angina pectoris**, is chest pain or pressure, usually caused by insufficient blood flow to the heart muscle (myocardium).[2] It is most commonly a symptom of coronary artery disease.[2]

- stable angina (more common) – attacks have a trigger (such as stress or exercise) and stop within a few minutes of resting
- unstable angina (more serious) – attacks are more unpredictable (they may not have a trigger) and can continue despite resting

Search Results for: angina

Related Terms: Stenocardia, Anginal syndrome, Angina pectoris, Ischemic heart disease - angina, AP - Angina pectoris, Cardiac angina

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Live Demo

Thank You